Dynamical Systems

CSC1111
Building Complex Computational Models

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Dynamical Systems – a short revision

- Dynamical systems - foundation of almost any kind of rule-based models of complex systems.

A dynamical system is a system whose state is uniquely specified by a set of variables and whose behavior is described by predefined rules.

- Some example include, population growth, a swinging pendulum, the behavior of “rational” individuals playing a negotiation game
- Can be described with difference or differential equations in discrete time steps or a continuous time line:

$$x_t = F(x_{t-1}, t) \qquad \frac{dx}{dt} = F(x, t)$$

  - where x_t or x is the state variable of the system at time t, which may take a scalar or vector value.
  - F is a function that determines the rules by which the system changes over time.
  - These are first-order versions of dynamical systems

Phase Space

- Behaviors of a dynamical system can be studied by using the concept of a *phase space*

A phase space of a dynamical system is a theoretical space where every state of the system is mapped to a unique spatial location.

- The number of state variables to uniquely specify the system's state is called the **degrees of freedom.** The degrees of freedom of a system equal the *dimensions* of its phase space.
- E.g. describing the behavior of a ball thrown upward in a frictionless vertical tube can be specified with two scalar variables, the ball's position and velocity. We can also create its phase space in two dimensions:

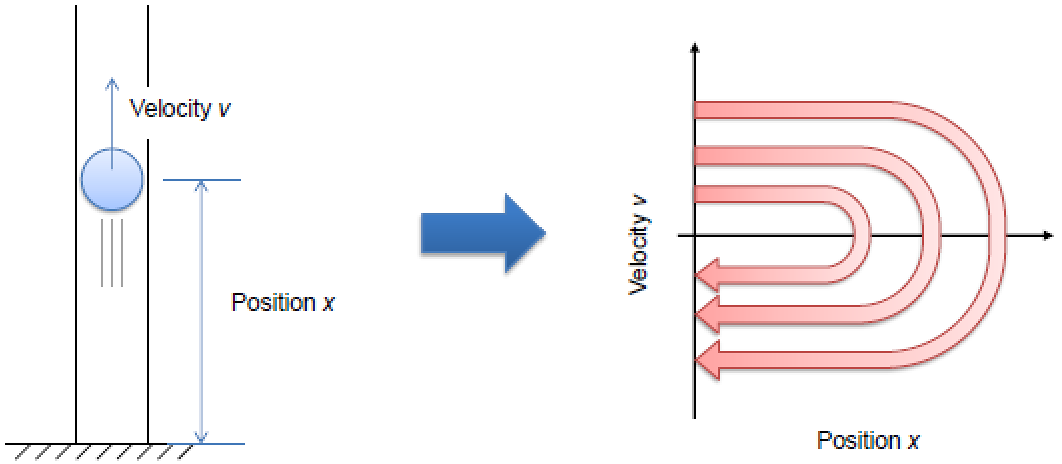


Figure: A ball thrown upward in a vertical tube (left) and a schematic illustration of its phase space (right).

- Drawing a phase space allows to visually represent the dynamically changing behavior of a system as a static trajectory. This provides intuitive, geometrical insight into the system's dynamics, hard just by looking at algebraic equations.

Phase Space – why important?

Phase space visualisations can help with several major learnings:

1) guess **system's state in a long run**.

- In deterministic dynamical systems, the future state is uniquely determined by its current state
- Trajectories of a deterministic dynamical system will never branch off in its phase space. We can visually inspect this phase space visualisation (they may diverge to infinity, converge to a certain point, or remain dynamically changing).

- Such a converging point or a region is called an **attractor**.
- The concept of **attractors** - important for understanding the self-organization of complex systems.

- Coming back to our definition from the early weeks - *Even if it may look magical, self-organisation of complex systems can be understood as a process where the system is simply falling into one of the attractors in a high-dimensional phase space*.

Phase Space – why important? –cont.

2) learn **how a system's fate depends on its initial state**.

- For each attractor, we can find *the set of all the initial states from which we will eventually end up*

falling into that attractor. This set is called the **basin of attraction** of that attractor. If there is more than one attractor in the phase space we can divide the phase space into *several different regions*.

- Such a "map" drawn on the phase space reveals *how sensitive the system is to its initial conditions*.

- If one region is dominating in the phase space, the system's fate doesn't depend much on its initial condition.
- If there are several regions that are equally represented in the phase space, the system's fate sensitively depends on its initial condition.

Phase Space – why important? –cont.

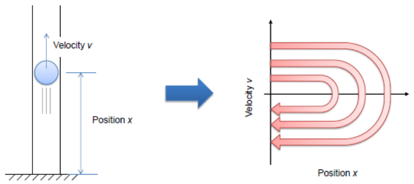
3) learn about **the stability of the system's states**.

- If we see that trajectories are **converging** to a certain point or area in the phase space, that means the system's state is **stable** in that area.
- If we see trajectories are **diverging** from a certain point or area, that means the system's state is **unstable** in that area.

- Knowing system stability is often extremely important to understand, design, and/or control systems in real-world applications (e.g. in bifurcation and chaos).

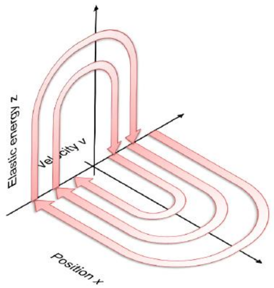
Phase Space – exercises

The bouncing ball example:



Ex.1. When the ball hits the floor in the vertical tube, its kinetic energy is quickly converted into elastic energy (i.e., deformation of the ball), and then it is converted back into kinetic energy again (i.e., upward velocity). Think about how you can illustrate this process in the phase space. Are the two dimensions enough or do you need to introduce an additional dimension? Then try to illustrate the trajectory of the system going through a bouncing event in the phase space.

- **Solution** You will need at least one more dimension to represent the amount of elastic energy stored in the ball. This dimension can be added to the phase space as a z-axis. A typical trajectory of the system going through bouncing events can be conceptually illustrated as follows:



It is assumed that part of the energy is lost at every bouncing event so that the trajectory is spiraling towards the origin.

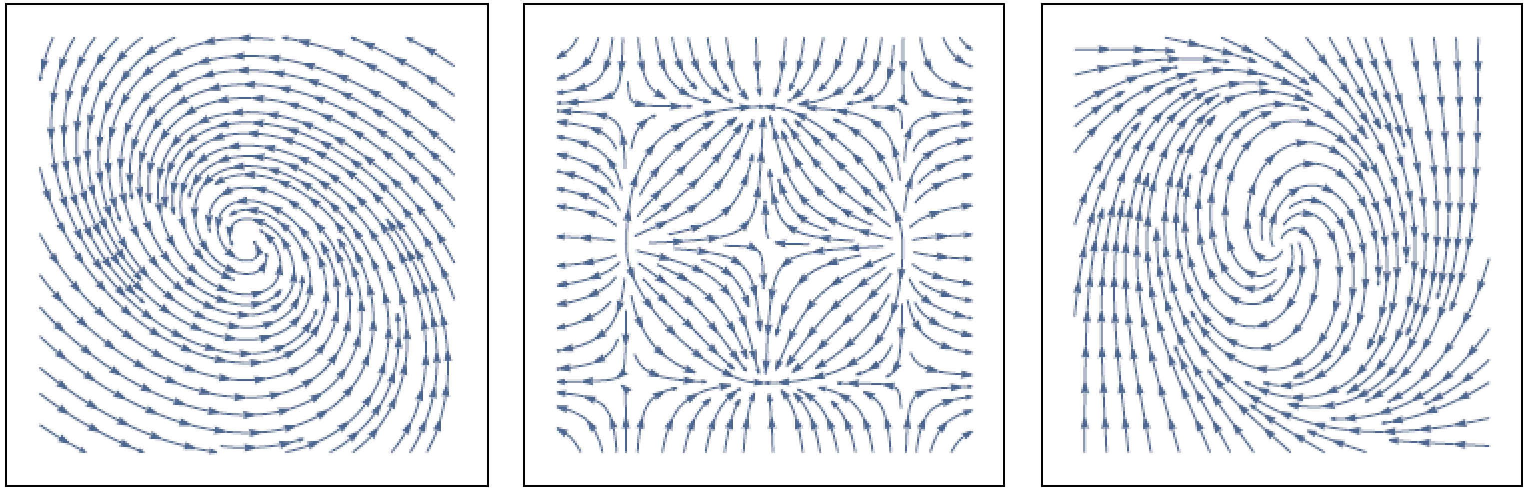
Ex.2. Where are the attractor(s) in the phase space of the bouncing ball example? Assume that every time the ball bounces it loses a bit of its kinetic energy.

- **Solution** The origin of the phase space is the only attractor in this case.

Phase Space – exercises

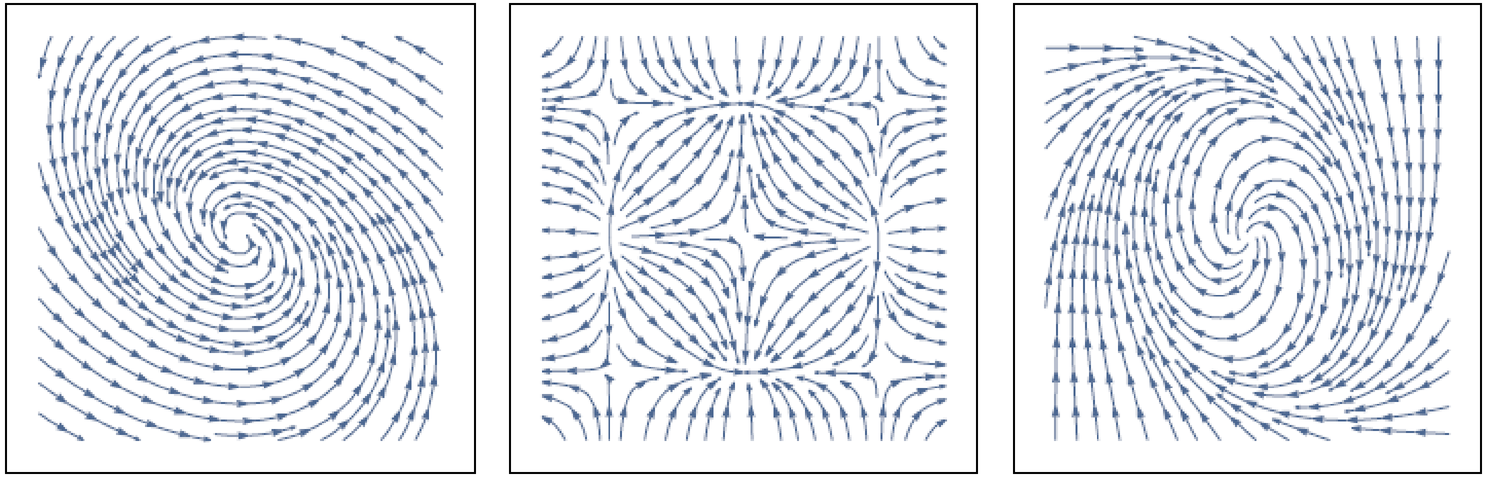
Ex.3. For each of the phase spaces shown below, identify the following:

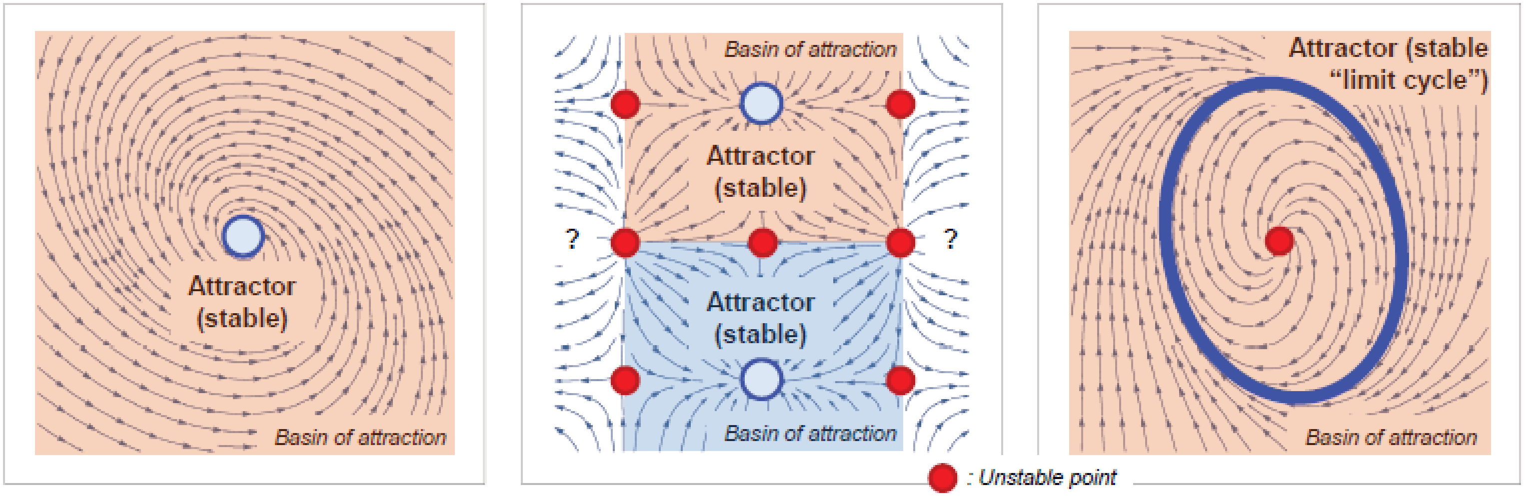
- attractor(s)
- basin of attraction for each attractor
- stability of the system's state at several locations in the phase space



Phase Space – exercises

Ex.3. Solution:





Types of systems – short examples

Dynamical systems can be split into several categories:

- **Linear** system - A dynamical equation whose rules involve just a linear combination of state variables (a constant times a variable, a constant, or their sum).
- **Nonlinear** system - Anything else (e.g., equation with squares, cubes, radicals, trigonometric functions, etc).

- **First-order** system - A difference equation whose rules involve state variables of the immediate past (at time $t - 1$) only.
- **Higher-order** system - Anything else.

- **Autonomous** system - A dynamical equation whose rules don't explicitly include time t or any other external variables.
- **Non-autonomous** system - A dynamical equation whose rules do include time t or other external variables explicitly.

- Some can be converted into another with additional state variables (e.g. Non-autonomous, higher-order difference equations can always be converted into autonomous, first-order forms).

Types of systems –exercise solution

Ex.4 Decide whether each of the following examples is

(1) linear or nonlinear,

(2) first-order or higher-order, and

(3) autonomous or non-autonomous.

1. $x_t = ax_{t-1} + b$
2. $x_t = ax_{t-1} + bx_{t-2} + cx_{t-3}$
3. $x_t = ax_{t-1}(1 - x_{t-1})$
4. $x_t = ax_{t-1} + bxt - 2^2 + c\sqrt{x_{t-1}x_{t-3}}$
5. $x_t = ax_{t-1}x_{t-2} + bx_{t-3} + \sin t$
6. $x_t = ax_{t-1} + by_{t-1}$, $y_t = cx_{t-1} + dy_{t-1}$

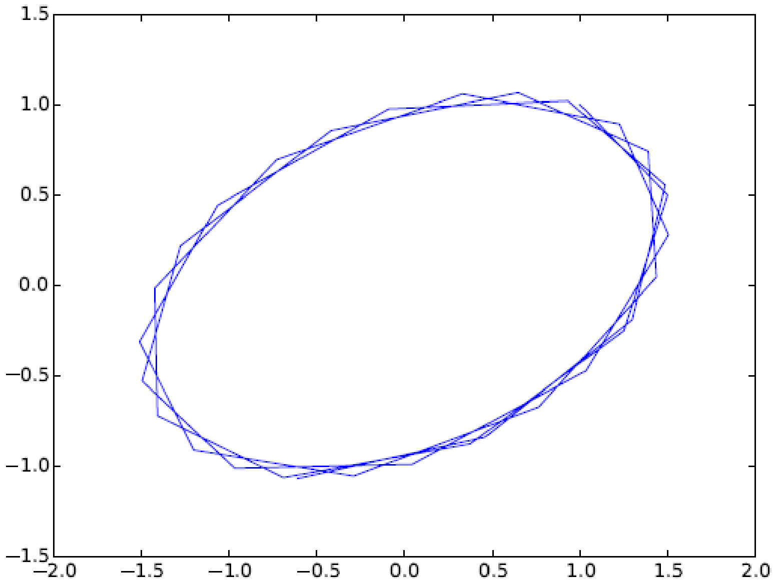
1. Linear, first-order, autonomous
2. Linear, higher-order, autonomous
3. Nonlinear, first-order, autonomous
4. Nonlinear, higher-order, autonomous
5. Nonlinear, higher-order, non-autonomous
6. Linear, first-order, autonomous

Dynamical systems - examples

<u>**Note**</u> – some examples on implementing dynamical systems in Python (you already did some in previous weeks) are uploaded in your Loop for week 11

- Linear dynamical systems can be used to show exponential growth/decay, periodic oscillation, stationary states (no change), or their hybrids (e.g., exponentially growing oscillation)
- You will practice how to visualise a phase space like the one below:

(we will see in bifurcations why that is important to understand)



Dynamical systems - examples

- **Ex. 5.** We discussed in previous weeks **predator/pray models**. Dynamical systems are heavily used for modelling those. Some general rules:
    - Prey grows if there are no predators.
    - Predators die if there are no prey.
    - The prey's death rate increases as the predator population increases.
    - The predators' growth rate increases as the prey population increases.

As an example, (…after a bit of maths which we do not need to do here…) these can mathematically be modeled as:

$$x_t = x_{t-1} + r x_{t-1}\left(1 - \frac{x_{t-1}}{K}\right) - \left(1 - \frac{1}{b y_{t-1} + 1}\right) x_{t-1}$$
$$y_t = y_{t-1} - d y_{t-1} + c x_{t-1} y_{t-1}$$

$$d_x(y) = 1 - \frac{1}{by + 1} \qquad r_y(x) = cx$$

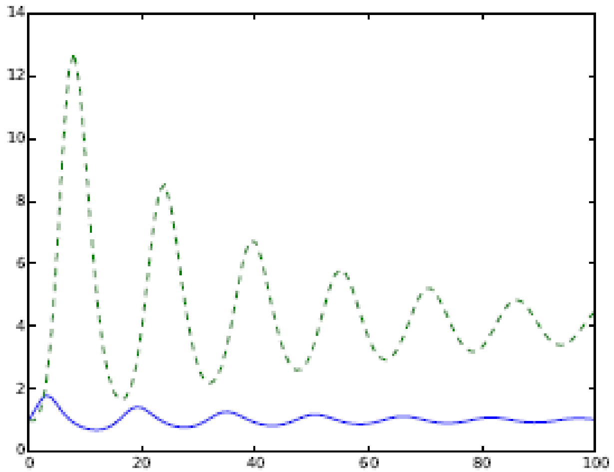
- Where x_t, y_t - logistic growth model for the prey. r_x is the growth rate of the prey, and d_y is the death rate of

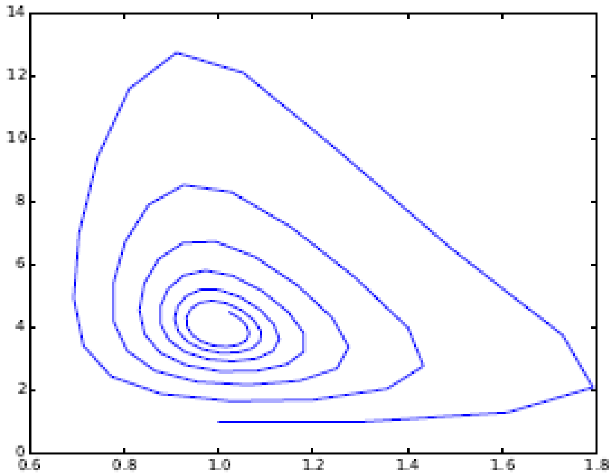
the predators $(0 < d_y < 1)$.

- The death rate of the prey should be 0 if there are no predators, while it should approach 1 (= 100% mortality rate!) if there are too many predators, $d_x(y)$. Similar for growth rate of the predators $r_y(x)$

Dynamical systems - examples

- **Ex. 5. – predator/pray models cont.**
- After varying simulation result *with* $r = b = d = c = 1$, $K = 5$ and $x_0 = y_0 = 1$ the system shows an *oscillatory behavior*.
- Its phase space visualization (right) indicates, it is not a harmonic oscillation seen in linear systems, but it is a nonlinear oscillation with distorted orbits.





Simulation results of the predator-prey model.

Left: State variables plotted over time.
Blue (solid) = prey, green (dashed) = predators.

Right: Phase space visualization.

- The model we have created is actually a variation of the **Lotka-Volterra model**, which describes various forms of **predator-prey interactions**. The Lotka-Volterra model is one of the most famous mathematical models of nonlinear dynamical systems that involves multiple variables and will be discussed again in later slides.
- For full code for this simulation check 'ds-predator-prey.py' in your Loop page under week 11.

Steady (Equilibrium) states

- When we analyse an autonomous, first-order discrete-time dynamical system like this one

$$x_t = F(x_{t-1})$$

one of the first things to find is its **equilibrium points** (or fixed points or steady states) (similar applies for continuous models).

- **Steady states** are those where the system can stay unchanged over time.
- **Equilibrium points** are important for both theoretical and practical reasons.
  - Theoretically, they are key points in the system's phase space, and help us to understand the structure of the phase space.
  - Practically, there are many situations where we want to sustain the system at a certain state that is desirable for us. In such cases, it is quite important to know whether the desired state is an equilibrium point, and if it is, stable or unstable.

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Bifurcations & Chaos

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Bifurcations: Introduction

- Important question when mathematically analysing a dynamical system is *how the system's long-term behavior depends on its parameters*?
- Mostly, we assume that a slight change in parameter values causes only a slight quantitative change in the system's behavior too, with the system's phase space unchanged.
- However, a slight change in parameter values can sometimes cause a drastic, qualitative change in the system's behavior, and the structure of its phase space can be topologically altered.
- This is called a **bifurcation**, and the parameter values at which a bifurcation occurs are called the **critical thresholds.**

Bifurcations

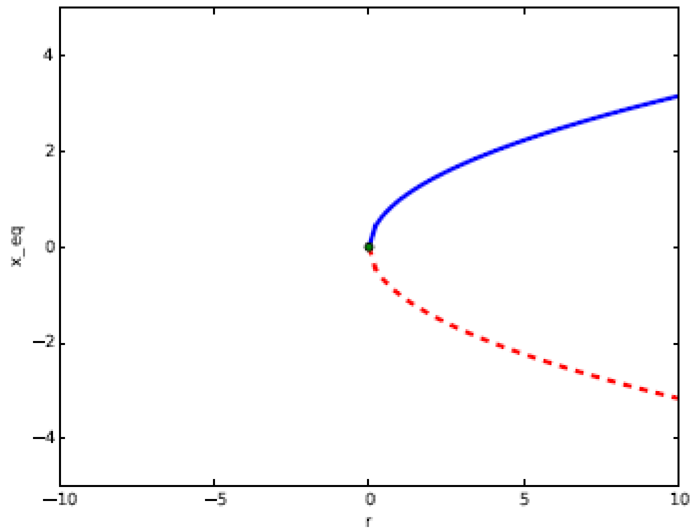
Def:

Bifurcation is a qualitative, topological change of a system's phase space that occurs when some parameters are slightly varied across their critical thresholds.

- Important in real-world systems as switching mechanisms e.g. excitation of neurons, catastrophic transition of ecosystem states, and binary information storage in computer memory.
- Two categories of bifurcations:
  - **local** bifurcation – change in the stability of equilibrium points. *Local*, because it can be detected and analysed only by using localised information around the equilibrium point.
  - **global** bifurcation – occurs when non-local features of the phase space, such as limit cycles, collide with equilibrium points in a phase space. Cannot be characterised just by using localised information around the equilibrium point.

Bifurcation diagram

- Bifurcation diagram showing a saddle-node for the following example: $\frac{dx}{dt} = r - x^2$

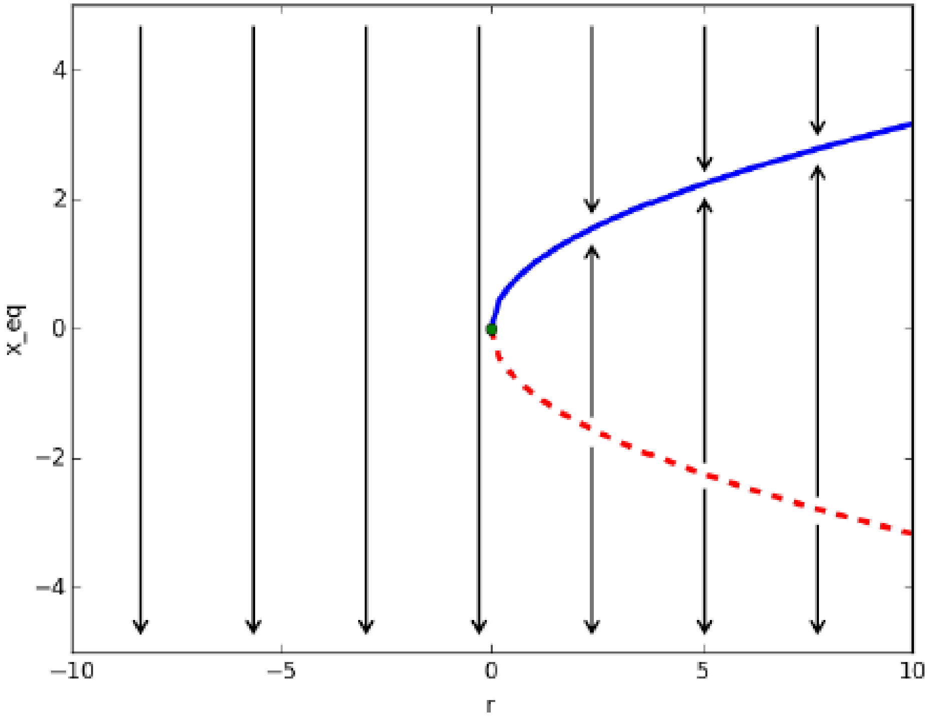


- The blue solid curve indicates a **stable equilibrium** point
- The red dashed curve indicates an **unstable equilibrium** point
- The green circle in the middle shows a **neutral equilibrium** point.

This type of bifurcation is called a **saddle-node bifurcation**, in which a pair of equilibrium points appear (or collide and annihilate), depending on which way you vary *r*.

- For full code for this simulation check ‘ds-bifurcation-diagram-saddlenode.py’ in your Loop page under week 11

Bifurcation diagram - interpretation



- How to interpret a bifurcation diagram?
- Each vertical slice of the diagram represents a phase space of the system for a particular parameter value.

Bifurcation diagram - example

- Imagine that we have a population model for a fish in a lake represented with: $\frac{dP}{dt} = \frac{-P^2}{50} + 2P$

where P - measured in thousands of fish and t is measured in years

- We want to investigate how some harvesting effect (i.e fishing) is influencing the amount of fish left in the lake. Let's write this as $$\frac{dP}{dt} = \frac{-P^2}{50} + 2P - 3h$$

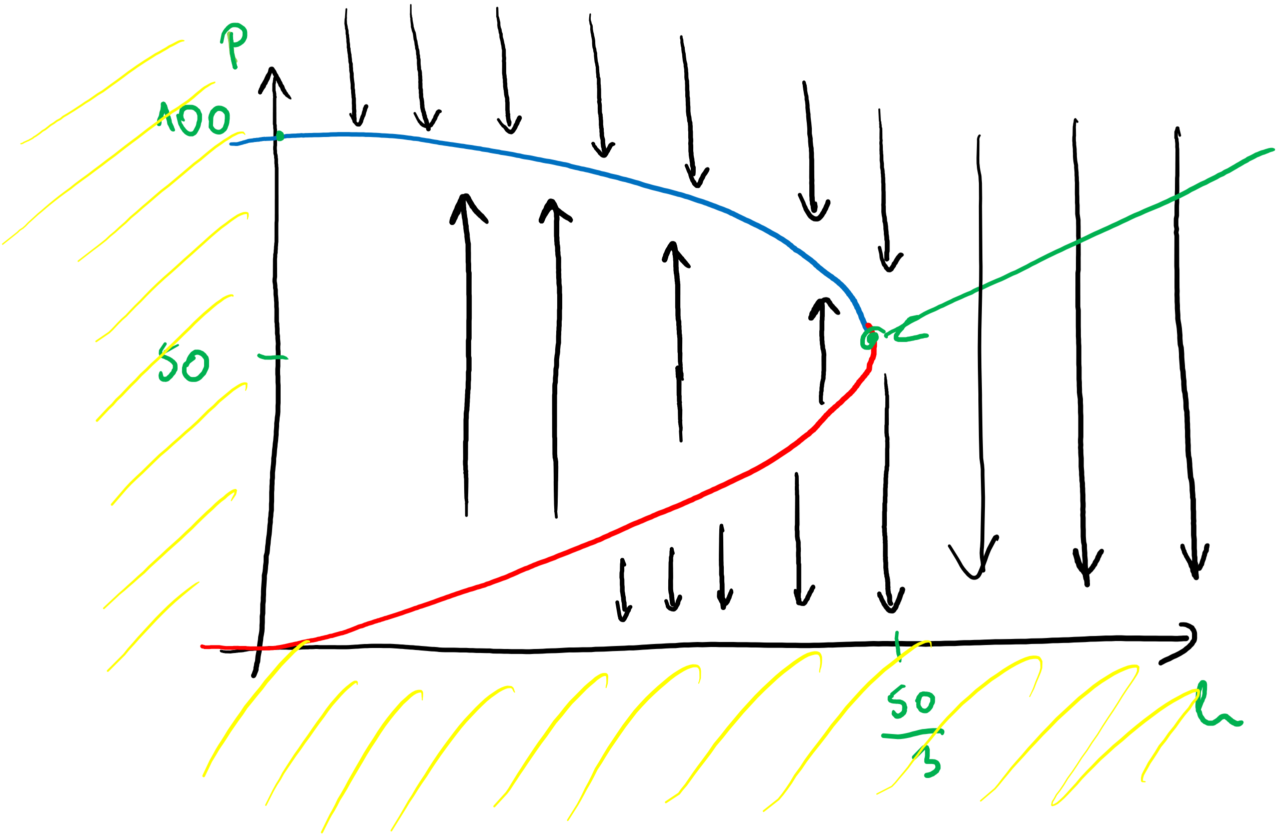
where h - the number of fishing licenses issued

- First, we find equilibrium of this equation and after solving this we get $h = \frac{50}{3}$ $P = 50\left(1 \pm \sqrt{1 - \frac{3h}{50}}\right)$

(values are actually not that important here but rather how we understand and read bifurcation diag).

- the diagram does not need to represent how h varies with P but rather how the overall system's state is behaving in critical points.

Bifurcation diagram - example



P - population of fish

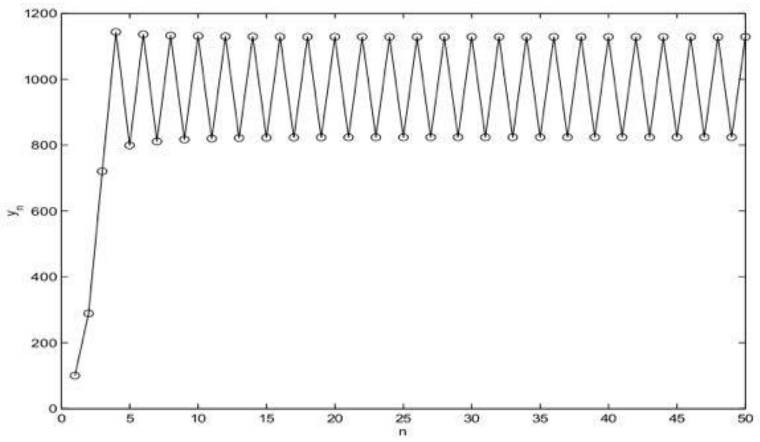
h - fishing licenses

Chaos theory

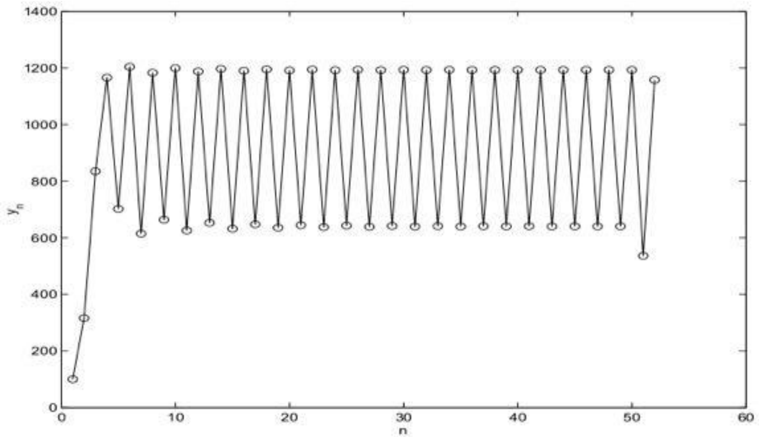
- *Chaos theory* is a mathematics theory dealing with nonlinear dynamical systems.
- The effect of *nonlinearity* is that the whole becomes something greater than just adding up the individual parts.
- This is due to feedback or multiplicative effects between the components.
- The meaning of *dynamical* is that the system changes over time based on its current state.
- Characteristic of chaotic systems is the presence of random states of disorder and irregularities.
- These are actually governed by underlying patterns and deterministic laws, **highly (extremely) sensitive** to initial conditions.

Bifurcations: Period-Doubling

- An example is a process of Period-Doubling i.e.: ***Bifurcation***



(a) k=2.1



(b) k=2.4

k – a growth rate of the system

Figure: The Two-Point Attractor

- In Figure above, there is a so-called **two-point *attractor***, where system settles down to alternating between two points.
- An **attractor** is a set of states of a dynamic physical system towards which it tends to evolve, whatever the system's starting conditions.
- An attractor is called **strange attractor** if it has a fractal nature i.e. it is not stable (if you zoom-in fractal indefinitely it will change all the time)

Bifurcation & Chaos

- A *bifurcation* is a period-doubling, a change from an *N*-point attractor to a *2N*-point attractor
- This occurs when the control parameter is changed.
- This is the parameter previously known as the growth rate *k*
- A *Bifurcation Diagram* is a visual summary of the succession of period-doubling produced as *r* increases.
- The next figure shows the bifurcation diagram of the *logistic map*, with *r* along the *x*-axis.

$$x_{n+1} = rx_n(1 - x_n)$$

- For each *r* value the system is first let settle down and then the successive values of x_n are plotted for a few hundred iterations.

Bifurcation & Chaos

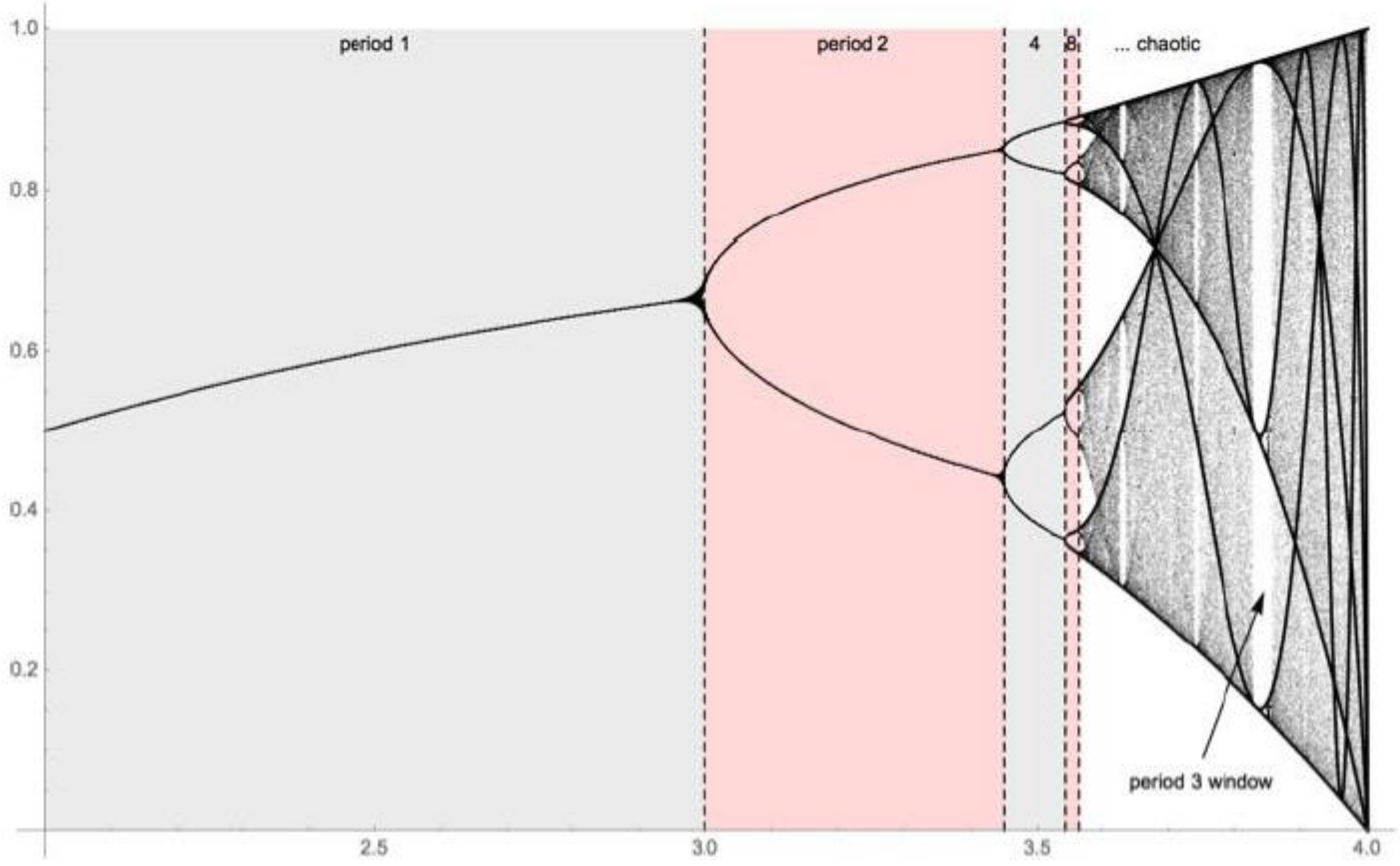


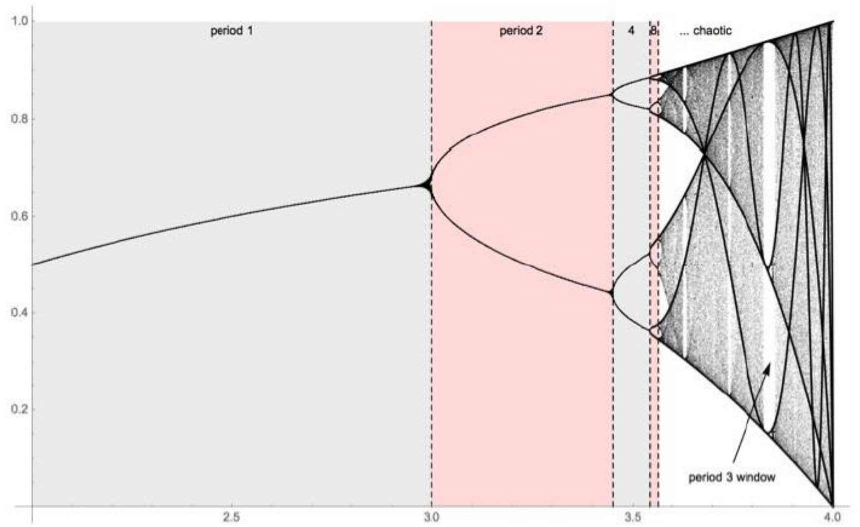
Figure: A Bifurcation Diagram

- On the diagram, are visible the various *N*-point attractors

Bifurcation & Chaos

- We can see, as *r* increases, the two cycle bifurcates into a four cycle at around $r \approx 3.45$.
- That happens repeatedly going through a so called *period doubling cascade*
  - We can see that infinite cascade actually completes around $r \approx 3.57$ and chaos ensues afterwards
  - There are, however, windows of periodic calm interspersed
  - The most prominent of these being the period 3 window around $r \approx 3.832$
  - A small change in *r* here can make a stable system chaotic
  - You can see this nicely at

https://www.geogebra.org/m/wQbHRgye



Bifurcation & Chaos

- A clear definition of 'Chaos' is hard to find, as many are to be found in the literature.
- Essentially it is that deterministic motion has the appearance of randomness.
- Here small changes to the initial condition will produce wildly different behaviour.
- This reinforces what was said earlier that simple models can have extremely complicated behaviour.
- Perhaps a better definition is that the present determines the future, but the *approx* present doesn't approximately determine the future.

Bifurcation & Chaos

Sensitivity to Initial Conditions

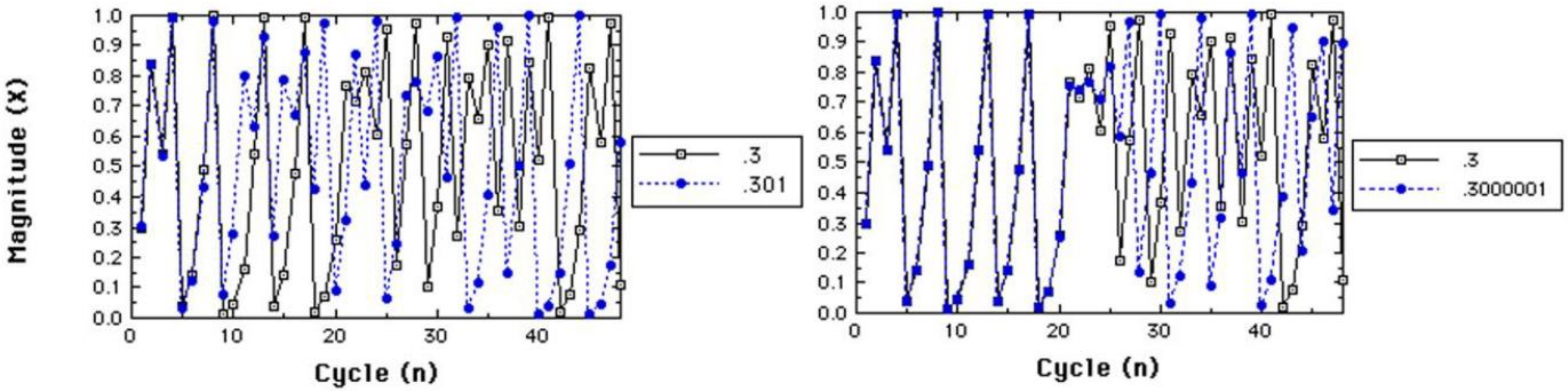


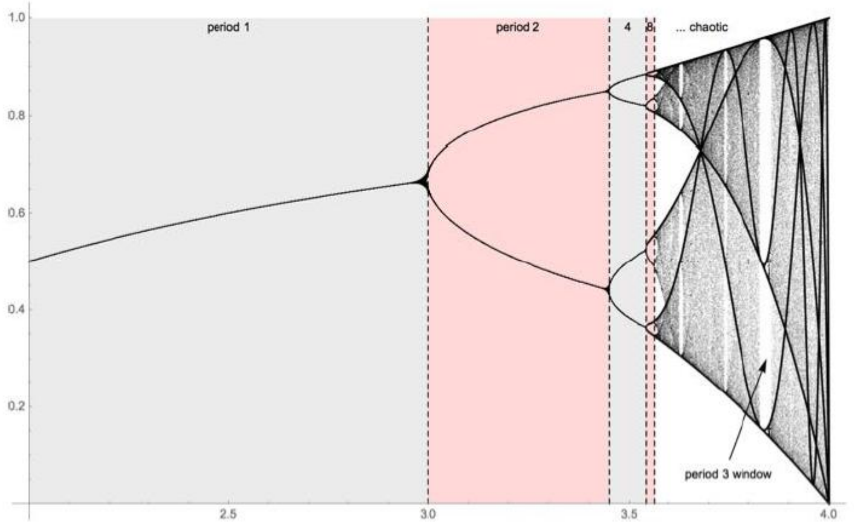
Figure: Logistic Growth: Sensitivity to Initial Conditions

- A factor of the chaotic region is sensitivity to initial conditions
- Can be seen in Figure above for values of *r > 3*

Bifurcation & Chaos

The Chaotic Region

- For *r > 3.5*, in the chaotic region, we see a **strange *attractor***
- The system oscillates forever around these, never repeating itself or settling into a steady state of behavior.
- It never hits the same point twice and its structure has a **fractal form**
- This implies the same patterns exist at every scale no matter how much you zoom into it.
- This fractality gives strange attractors great detail and complexity.
- We can see this on the next page.



Bifurcation & Chaos

The Chaotic Region

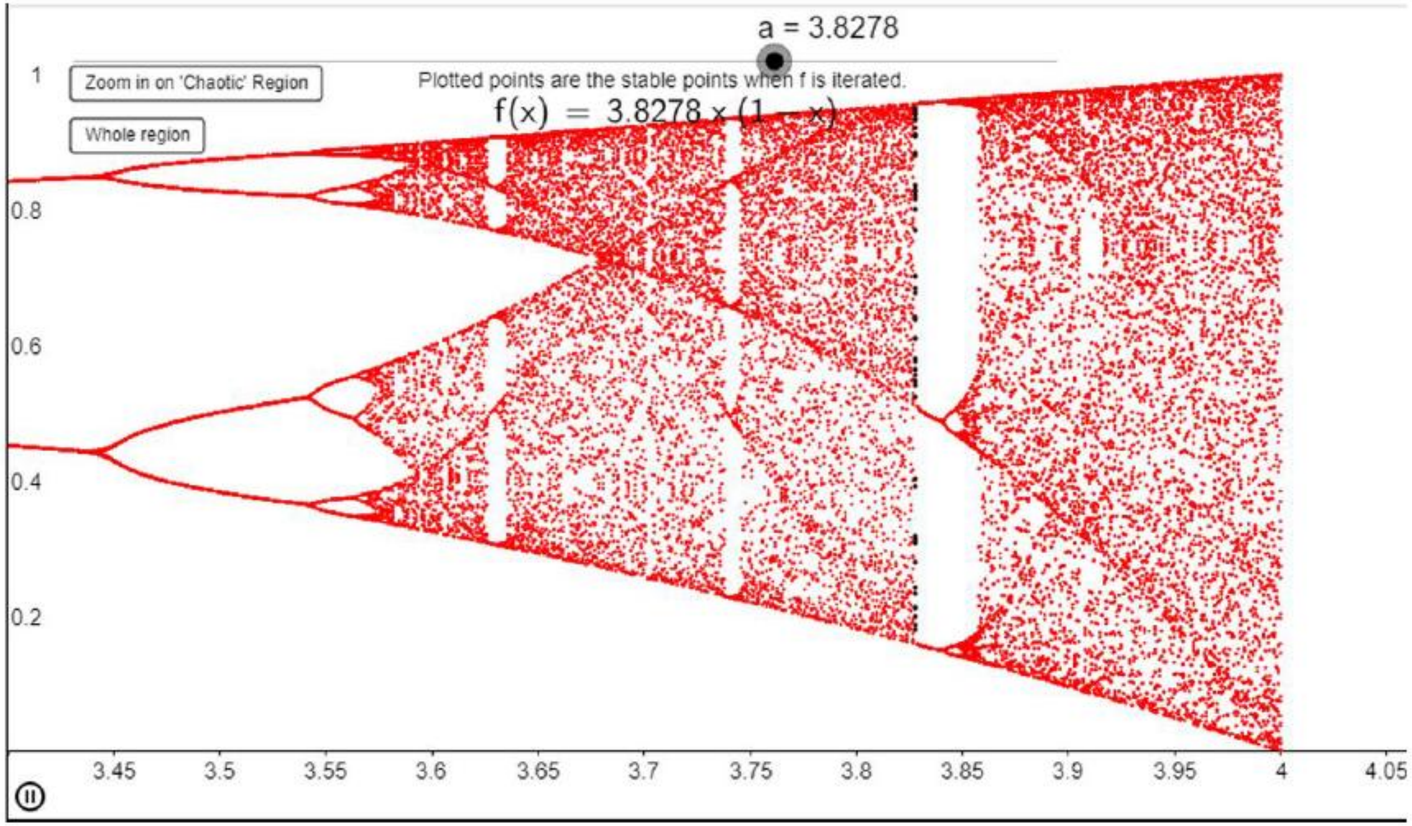


Figure: The Chaotic Region and Strange Attractors

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Chaos in higher dimensions FYI only

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Chaos in Higher Dimensions

- Up to now, we have seen chaos in one dimension in terms of the logistic map
- In the next part, we concentrate on examples of Chaos in higher dimensions
- For such systems, it is frequently easier to visualize what happens in ***Phase Space***
- Or, in other words, with one variable plotted in terms of another, without time explicitly displayed.
- So with a system in two or more dimensions
    - its current state is the current values of its variables, and
    - treated as a point in phase (state) space, and
    - we refer to its *trajectory* or *orbit* in time.

The Logistic Map, Generalised

- Consider again the Logistic map:

$$x_{n+1} \;=\; r x_n (1 - x_n)$$

- Multiplying the right side out:

$$x_{n+1} \;=\; r x_n - r x_n^2$$

- And, generalising, gives:

$$x_{n+1} \;=\; a x_n - b x_n^2$$

- Here, *a, b* are growth, suppression parameters, respectively.

The Lotka-Volterra Map

- Firstly, we consider a 2D system chaotic system with Rabbits/ Wolves
- This is an example of a *Lotka-Volterra Map*

$$\begin{aligned} x_{k+1} &= x_k + ax_k + -dx_k y_k \\ y_{k+1} &= y_k + -by_k + ex_k y_k \end{aligned}$$

- Here:
  - *x, y* represent the number of prey, predator
  - *a, b* is the growth rate of prey, predator independent of each other
  - *e* is the growth rate of the predator per prey consumed,
  - *d* is the rate of consumption of prey per predator.

The Lotka-Volterra Map

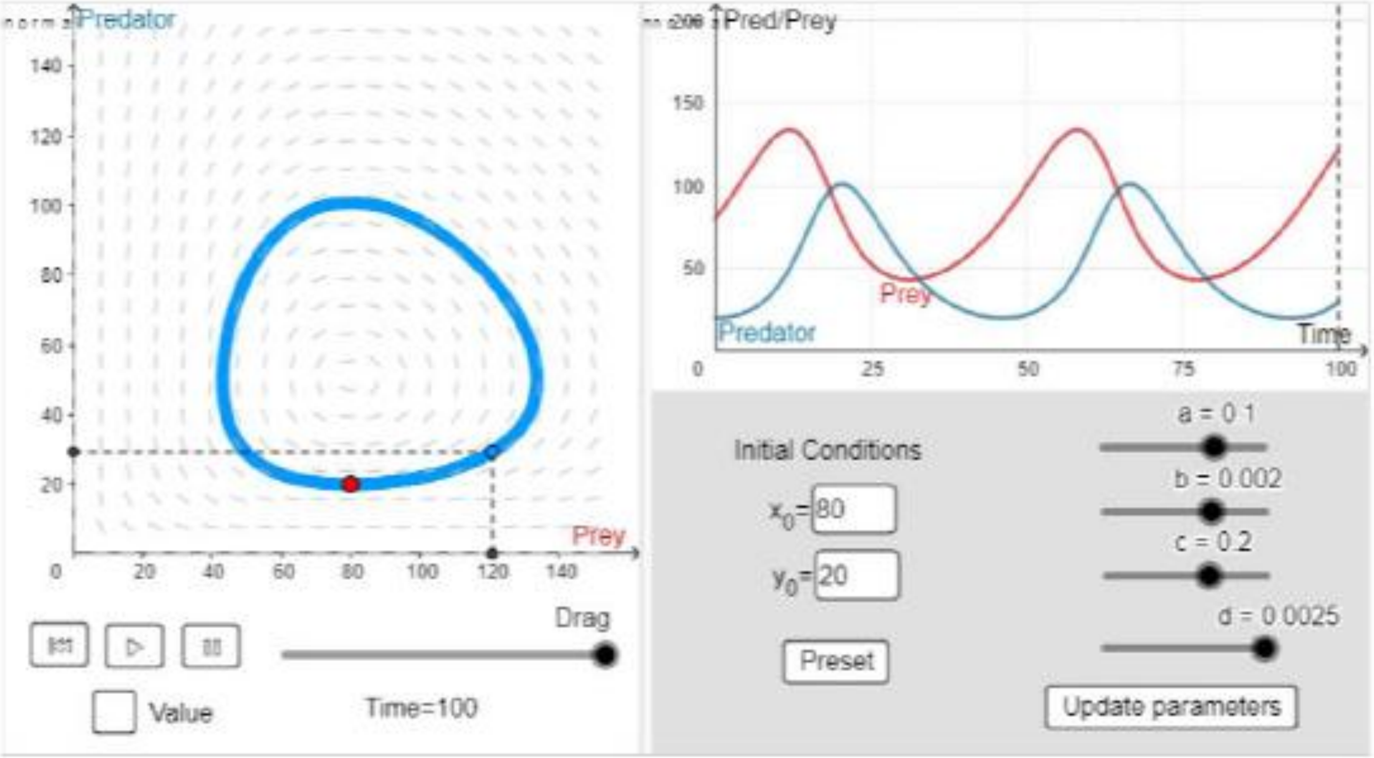

Figure: The Lotka-Volterra Map

- This is a 2D system where 2 species grow, one at the other's expense.
- Predator numbers always lag those of prey as the cycle tracks anti-clockwise, with parameter values as shown
- An attractor forming a loop like this is called a **limit cycle**

The Lotka-Volterra Map

- As may be seen neither equilibrium point is stable.

– the predator and prey populations seem to cycle endlessly without settling down quickly.

- While this cycling has been observed in nature, it is not overwhelmingly common.
- It appears that Lotka-Volterra by itself or in its standard form is not sufficient to model many predator-prey systems.
- For this reason it is necessary to further parameterise the model.
- Generalizing eqn

$$\begin{aligned} x_{k+1} &= x_k + a x_k + -d x_k y_k \\ y_{k+1} &= y_k + -b y_k + e x_k y_k \end{aligned}$$

to account for logistic growth of prey and harvesting of predators, we get:

- With the phase plane plot:

$$\begin{aligned} x_{k+1} &= x_k + a x_k + -c x_k^2 - d x_k y_k \\ y_{k+1} &= y_k + -b y_k + e x_k y_k - h \end{aligned}$$

The Lotka-Volterra Map

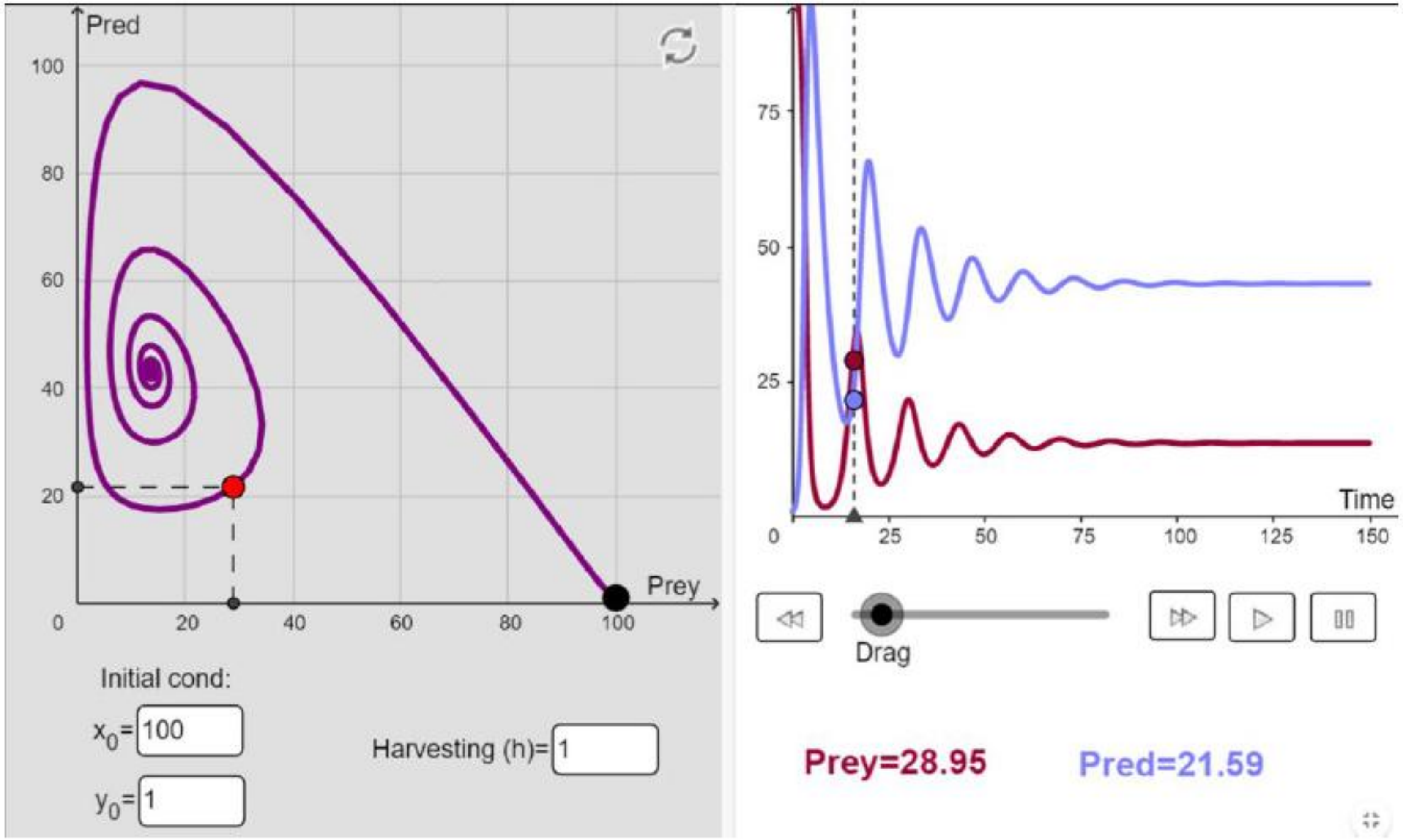

Figure: The Lotka-Volterra Map Revisited

The Hénon Map

- The next example of a map is the so-called Hénon Map
- A logistic map maps a 1D real interval [0 . . . 1] onto itself
- Whereas A Hénon map maps a point (x_n, y_n) in the 2D real plane onto a new point (x_{n+1}, y_{n+1}):

$$\begin{aligned} x_{n+1} &= 1 - a x_n^2 + y_n \\ y_{n+1} &= b x_n \end{aligned}$$

- As may be seen from above eqn the map depends on two control parameters, *a* and *b*
- For the classical Hénon map $a = 1.4$, $b = 0.3$ for which the map is chaotic.
- A nice app for varying a, b and visualising the map can be seen at:

https://experiences.mathemarium.fr/Henon-s-attractor.html

The Hénon Map

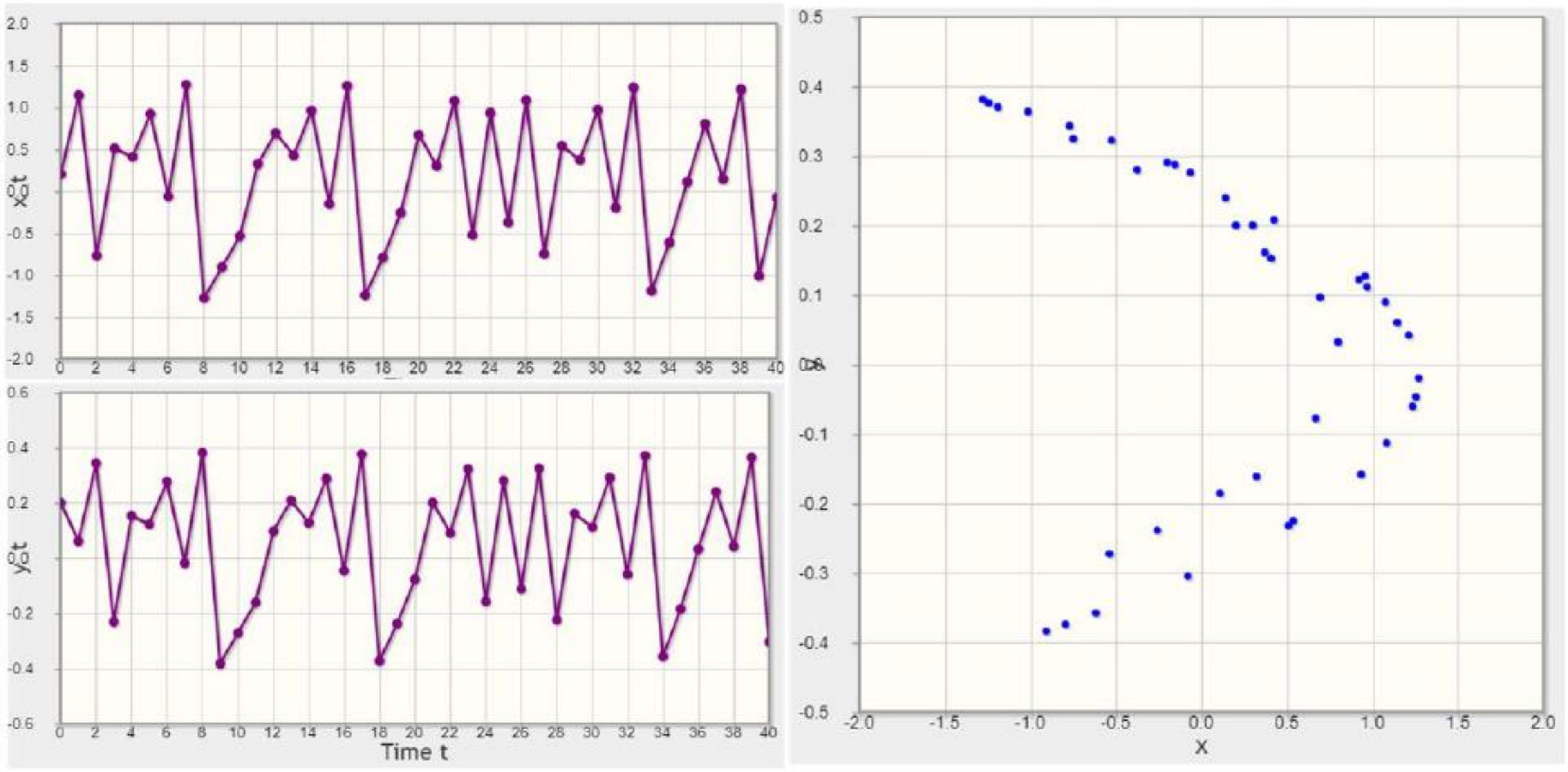

Figure: The Henon Map

The Hénon Map

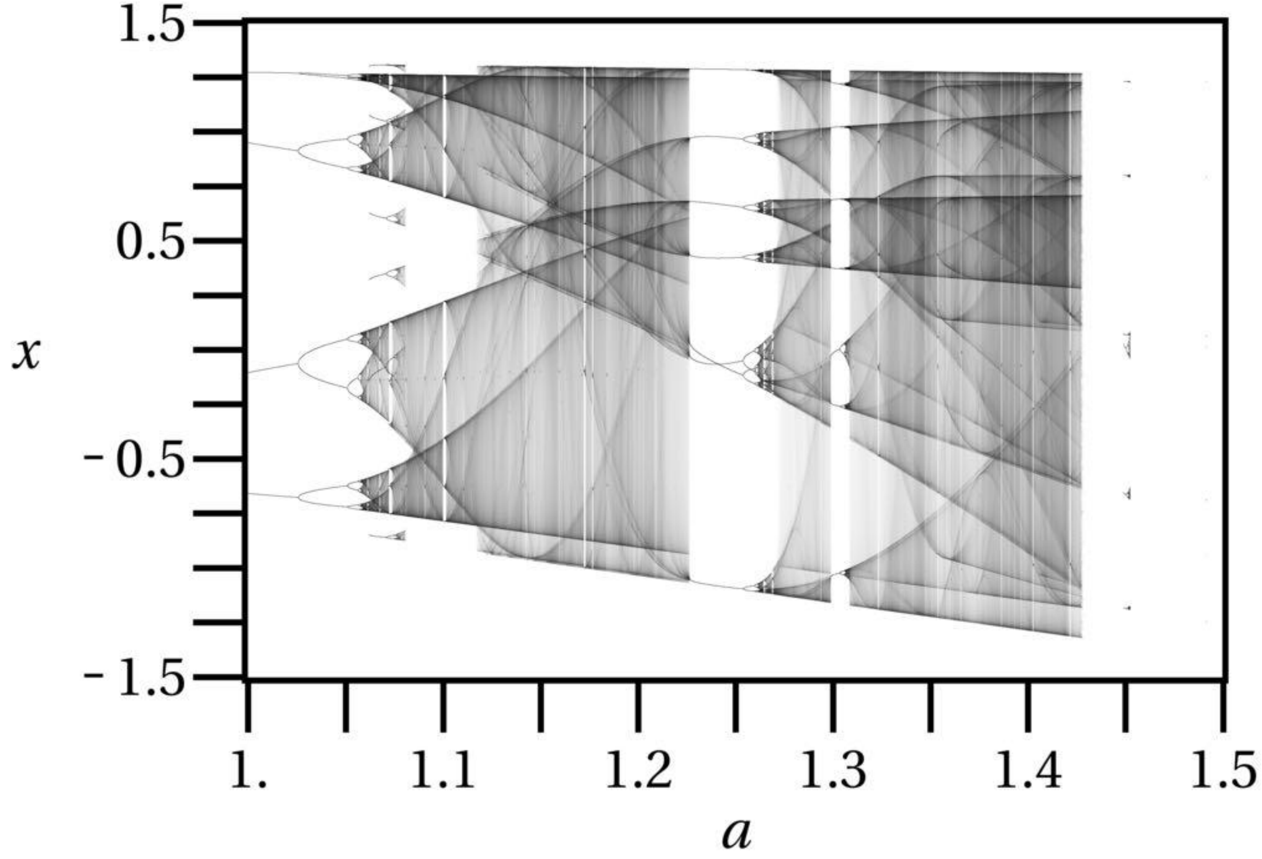

Figure: The Henon Bifurcation Diagram (by Jordan Pierce)

- In Fig. above darker regions denote higher density in state space diagram

Chaos in 3D: The Lorenz Equations

Background to the Lorenz Equations

- In 1960’s American Meteorlogist, Edward Lorenz was studying Atmospheric Convection
- In attempting to simplify the Navier-Stokes Equations*, he reduced system to an ODE system
- He wanted to explore the fundamental limitations to weather forecasting
- During this he stumbled upon a chaotic dependence on initial conditions, which he termed ‘The Butterfly Effect’
- Originally, this was ‘one flap of a sea gull’s wings would be enough to alter the course of the weather forever’
- Later assumed the more familiar, punchier title of ‘Does the flap of a butterfly’s wings in Brazil set off a tornado in Texas?’

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* Non-Linear PDEs governing Fluid Mechanics

Chaos in 3D: The Lorenz Equations

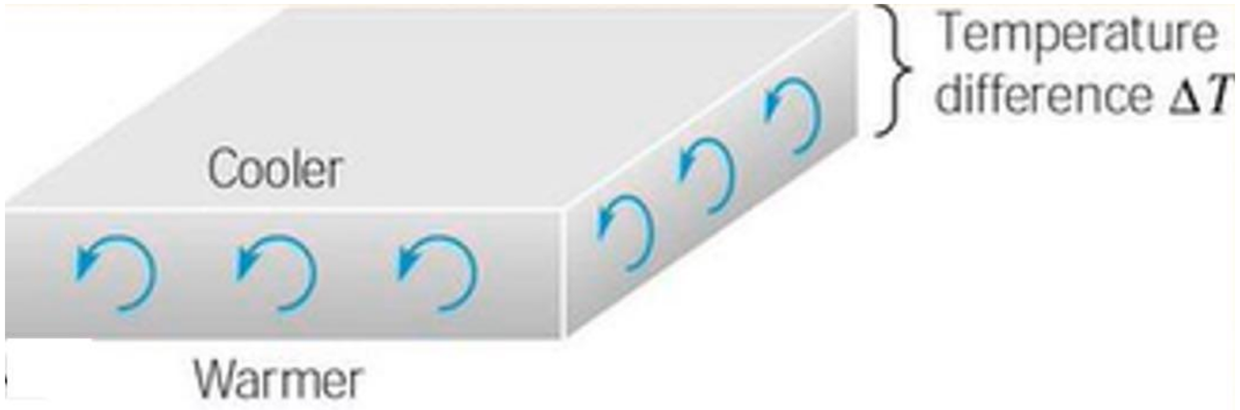

Figure: Schematic Diagram of System Lorenz was Studying

- We treat the atmosphere as a fluid layer, heated at bottom, cooled at top.
- Here, we have the Navier-Stokes equations for fluid motion.
- With flow is driven by thermal convection/ diffusion.

Chaos in 3D: The Lorenz Equations

- The equations of motion are then:

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y-x)\\ \frac{dy}{dt} &= x(\rho-z) - y\\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

- Here we have a number of dimensionless quantities:
  - *x, y, z* are dimensionless displacement variables
  - Prandtl Number, σ , the ratio of momentum diffusivity to thermal diffusivity
  - Rayleigh Number*, ρ, a measure of natural convection
  - ß, relating to certain physical dimensions of the layer itself

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*low for laminar, high for turbulent flow

Chaos in 3D: The Lorenz Equations

- Assumptions of the Lorenz Equations:
  1. Nonlinearity - the two nonlinearities are xy and xz
  2. Symmetry - Equations are invariant under $(x; y) \rightarrow (-x, -y)$.

  Hence if $(x(t); y(t); z(t))$ is a solution, so is $(-x(t); -y(t); z(t))$
  3. Volume contraction - The Lorenz system is dissipative i.e. volumes in phase-space contract under the flow

- The equations describe temperature differences and air movement
- If the layer is thin, or temperature difference ΔT is small, then T varies linearly with altitude but no significant fluid motion results.
- If ΔT is large enough, then a steady convection motion results.
- As ΔT increases, a more complex and turbulent motion eventually ensues.

Lorenz Equations: Stability

- In order to determine the regions of stability of the Lorenz Equations, we must find the *Critical Points*
- These are the points for which the derivatives of Eqn are instantaneously zero.

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y-x) \\ \frac{dy}{dt} &= x(\rho-z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

- i.e. rate of change in *x, y, z* directions is instantaneously zero.
- 3 points result:

$$\begin{aligned}&(0,0,0), \\ &(\beta\sqrt{\rho-1}, \beta\sqrt{\rho-1}, \rho-1), \\ &(-\beta\sqrt{\rho-1}, -\beta\sqrt{\rho-1}, \rho-1)\end{aligned}$$

- The latter pair of critical points is only stable for $\rho < \sigma\frac{\sigma+\beta+3}{\sigma-\beta-1}$
- For simplicity, usually assume *σ = 10* and *ß = 8/3* so leaves the behaviour a function of *ρ* only

Lorenz Equations: Stability

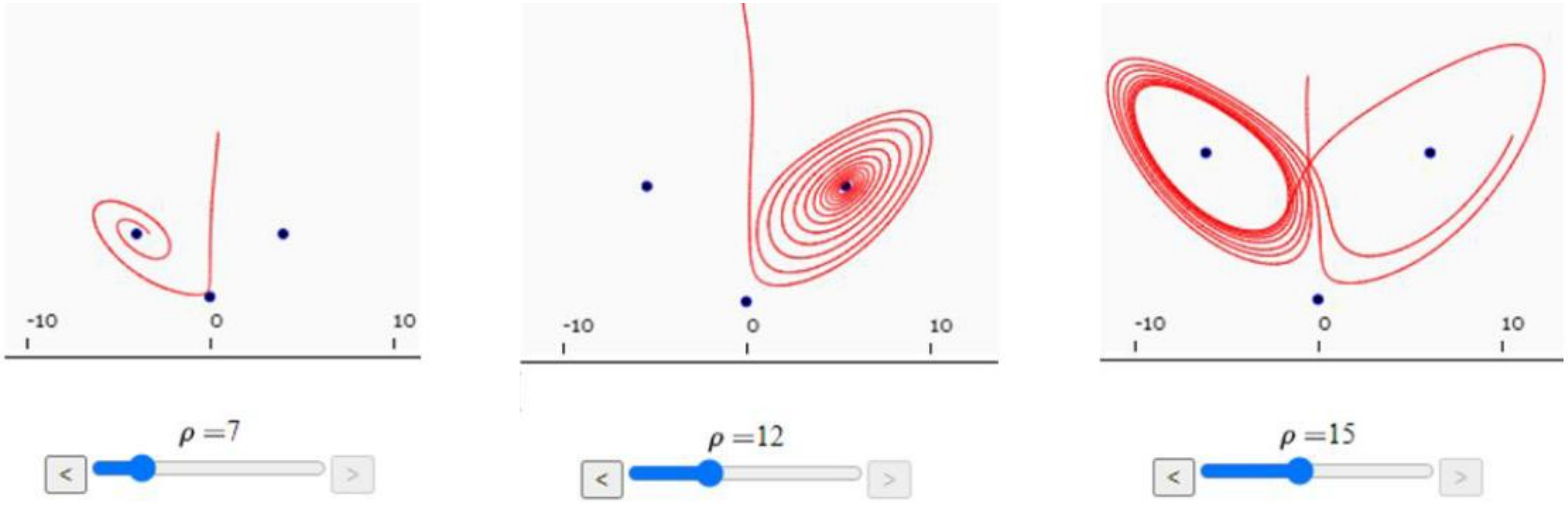

Figure: Lorenz z-domain plots for various ρ

- For different ρ, a number of regions of stability can be seen:

1. For small ρ, the system is stable and evolves to one of two fixed point attractors (first two plots of Figure)
2. For $\rho \geq 14$, system forms unstable limit cycles, oscillating between attractors unpredictably (Figure right)
3. This continues until $\rho \geq 25$, where fixed points attractors become repulsors, the trajectory repelled by them in a very complex way.

- Plot is from app by Hendrik Wernecke https://itp.uni-frankfurt.de/~gros/Vorlesungen/SO/simulation_example/#simulation

Lorenz Equations: Stability

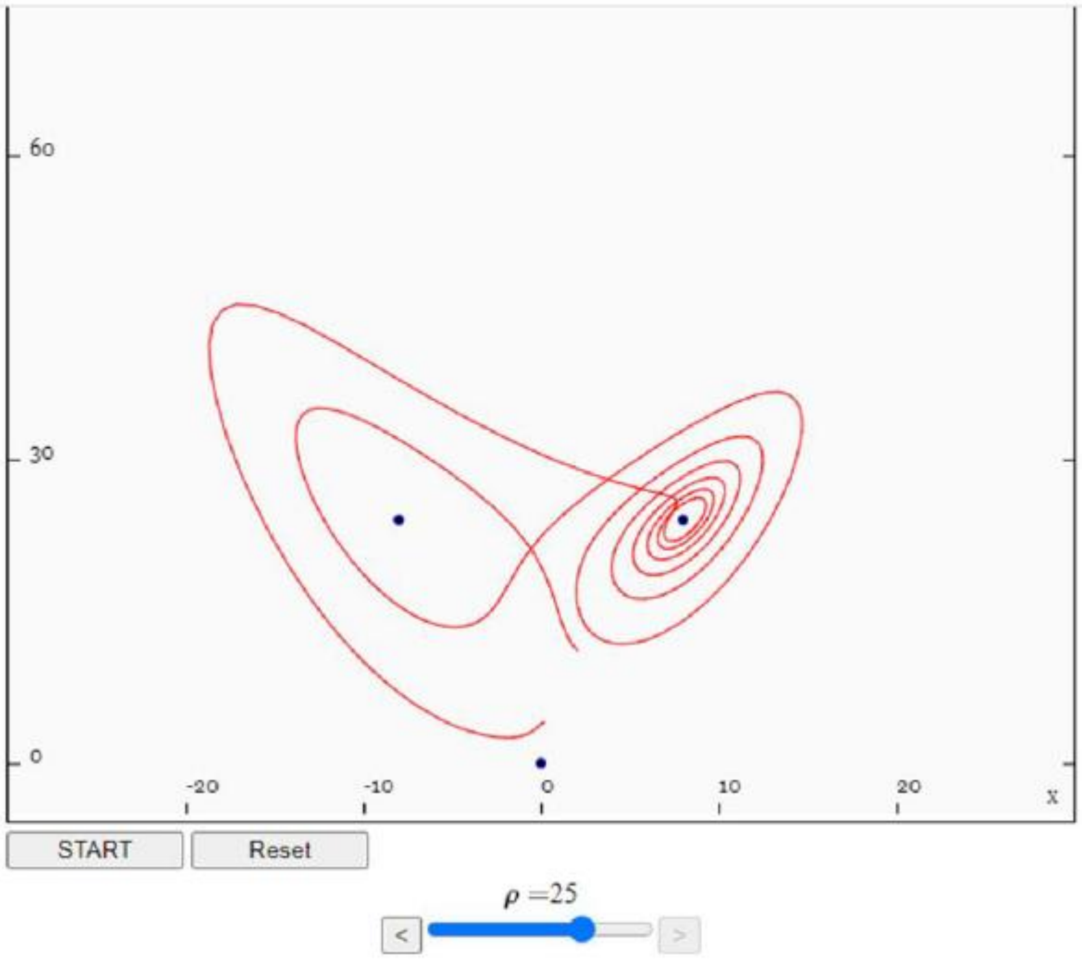

Figure:
A Lorenz Butterfly Plot

For more nice links for Lorenz attractor plots check:
https://www.t-ott.dev/2022/02/24/lorenz-attractor
https://experiences.mathemarium.fr/Lorenz-attractor (for 3D)

Lorenz Equations: 3D plots

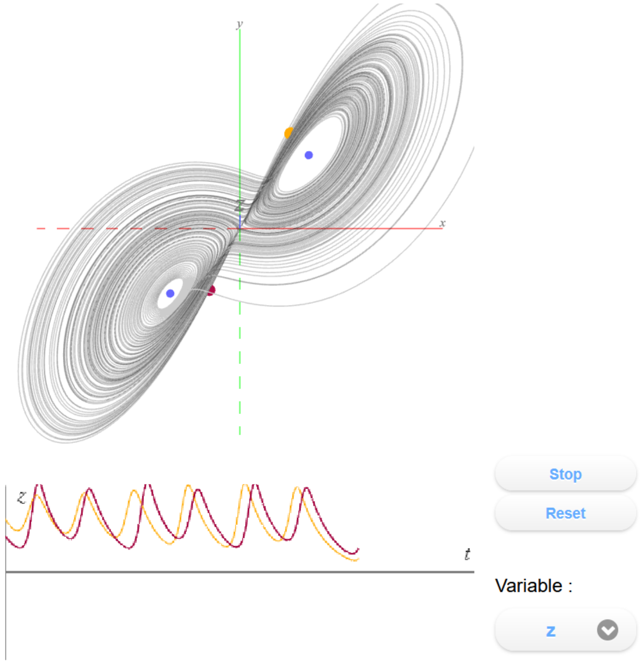

Figure:
3D Lorenz Plot

For more good links for Lorenz attractor plots check in labs:
https://experiences.mathemarium.fr/Lorenz-attractor (for 3D)

- We can flip diagrams around to see how the two points are behaving
- We can see what is happening for the two strange attractors how they have a very fine response to the initial conditions